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Semantic Web

LexNet: A Linked Open Data Approach to Trilingual Vocabulary Learning

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IT6390E - Group 20

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Abstract

Vietnamese learners of English and Chinese can link words of the two languages through their Vietnamese translations, but this leads to wrong matches when a translation has several meanings: through *đường*, English *sugar* reaches 路 ‘road’ as well as 糖 ‘sugar’. LexNet, a vocabulary resource for these learners, instead links the senses of 759 English, Vietnamese and Chinese words to 236 shared concepts, so *sugar* reaches only 糖. Built on OntoLex-Lemon, SKOS and LexInfo, it attaches levels, registers and usage examples to these senses, marks confusable words and breaks Chinese words down into characters and radicals. Its data is published as one openly licensed RDF file with dereferenceable IRIs and links to Wikidata items and lexemes, which makes it five-star Linked Open Data. A SPARQL endpoint answers nine competency questions, partly through federated queries to Wikidata, and a web application presents the data to learners. The reasoner HermiT finds the OWL 2 DL ontology and the data consistent, infers polysemous words and the reverse of symmetric relations, and detects all 12 faults added to the data one at a time. A query by radical shows that all five words with a character under 犭 name mammals, but only six of the nine under 水 ‘water’ concern liquids, so a radical hints at the meaning of a word without determining it.

1. Introduction

Nation [26, ch. 2] groups nine aspects of knowing a word under form, meaning and use. Besides the link between form and meaning, they include associations with other words and constraints on use such as register. Learner resources for English mark register and proficiency level, sometimes for one meaning of a word rather than for the word as a whole. In the Oxford 3000 and 5000 list [28], *bank* is A1 in its financial sense and B1 in the sense of land beside a river.

For Chinese, the HSK 3.0 syllabus [8] assigns a level to each word it lists, and Chinese characters carry information of their own. UniHan [31] classifies 猫 ‘cat’, 狗 ‘dog’ and 猪 ‘pig’ under radical 94 犬 ‘dog’, which takes the form 犭 in these characters. Li et al. [22] note that 犭 refers to the category “mammal”. A radical can thus group words by meaning.

The CEFR Companion Volume treats the languages a person knows as one inter-related repertoire [11, p. 30]. For a Vietnamese speaker who learns both English and Chinese, the three languages form such a repertoire. Yet the Oxford resources and the HSK syllabus cover English and Chinese separately. A vocabulary resource that follows the CEFR view would instead connect the three languages.

Words that translate each other share a meaning rather than a form, so such a connection has to go through meanings. One way to connect English and Chinese is through their Vietnamese translations, using a Vietnamese–English and a Vietnamese–Chinese dictionary. This route goes through Vietnamese words rather than meanings, and a Vietnamese word may have several meanings. Vietnamese *đường* means both ‘sugar’ and ‘road’, so the route from English *sugar* through *đường* reaches Chinese 路 ‘road’ as well as 糖 ‘sugar’. Bond and Foster [5] point to the same ambiguity when bilingual dictionaries are linked through a common language. Their Open Multilingual Wordnet avoids it by linking languages through the synsets of the Princeton WordNet [25], which stand for shared meanings.

However, the levels, registers and radicals above come from resources that do not use synsets or any other shared identifiers for meanings. Bringing their information together around shared meanings needs identifiers that all the data can refer to. Linked Data provides a way to create

such identifiers and to link them across datasets. Each thing is named by an HTTP IRI that returns a description in RDF when it is looked up, and descriptions link to things in other datasets [3, 4]. Linked Data released under an open licence that allows free reuse is called Linked Open Data [3]. Wikidata [32] is an example that represents concepts as items and, in its lexicographical data [35], words as lexemes. Other datasets can link their own concepts and words to these items and lexemes.

Representing words and meanings as Linked Data calls for a model of the lexicon. OntoLex-Lemon [10, 23] separates a word from its senses and from the concepts they express, so words of different languages can share a concept through their senses. Information that holds for a word in only one of its meanings, such as the two levels of *bank*, can then be attached to the sense.

LexNet is a vocabulary resource for Vietnamese learners of English and Chinese. It follows the approach above and organises the vocabulary of the three languages around shared meanings, modelled as language-independent concepts. Every word sense points to one of these concepts, so translations and synonyms follow from the concepts they share instead of being listed pair by pair. Relations between concepts (hypernymy, meronymy and antonymy) record some of the associations between words. A learner layer records levels, register and usage examples on word senses and marks confusable words. A script layer breaks Chinese words down into characters and radicals.

The vocabulary covers everyday topics such as animals, family, food, numbers and time. LexNet does not record what an individual learner knows. The aim is to publish LexNet as five-star Linked Open Data [1] and to make it accessible through a SPARQL endpoint and a web application. The work covers an ontology for the concepts and the two layers, the conversion of the data into RDF under an open licence with a dereferenceable IRI for every resource, and links from concepts and words to Wikidata items and lexemes.

2. Ontology Design

The ontology reuses three standard vocabularies. OntoLex-Lemon [10] models words, their senses and the concepts they express. SKOS [24] models concept schemes such as levels and topics, and LexInfo [9] provides parts of speech. LexNet's own terms use the prefix `lexnet::`. The ontology stays within OWL 2 DL [33], so that a reasoner can check the data against it (Section 6). The main classes and properties are shown in Figure 1.

2.1 Requirements

Following Noy and McGuinness [27], the design starts from competency questions that the ontology must be able to answer. Section 5 shows how SPARQL queries answer them. The questions are as follows:

- CQ1 Which English, Vietnamese and Chinese words express a concept, and at which level?
- CQ2 Which words of the same language share a meaning with a given word, and in which register is each one used?
- CQ3 Which meanings does a word have, and what are the level and a usage example of each?
- CQ4 Which concepts are kinds, parts or opposites of a given concept?
- CQ5 Which words of a topic are at or below a given level?
- CQ6 Which words can be confused with a given word because they have close meanings, look alike or sound alike?

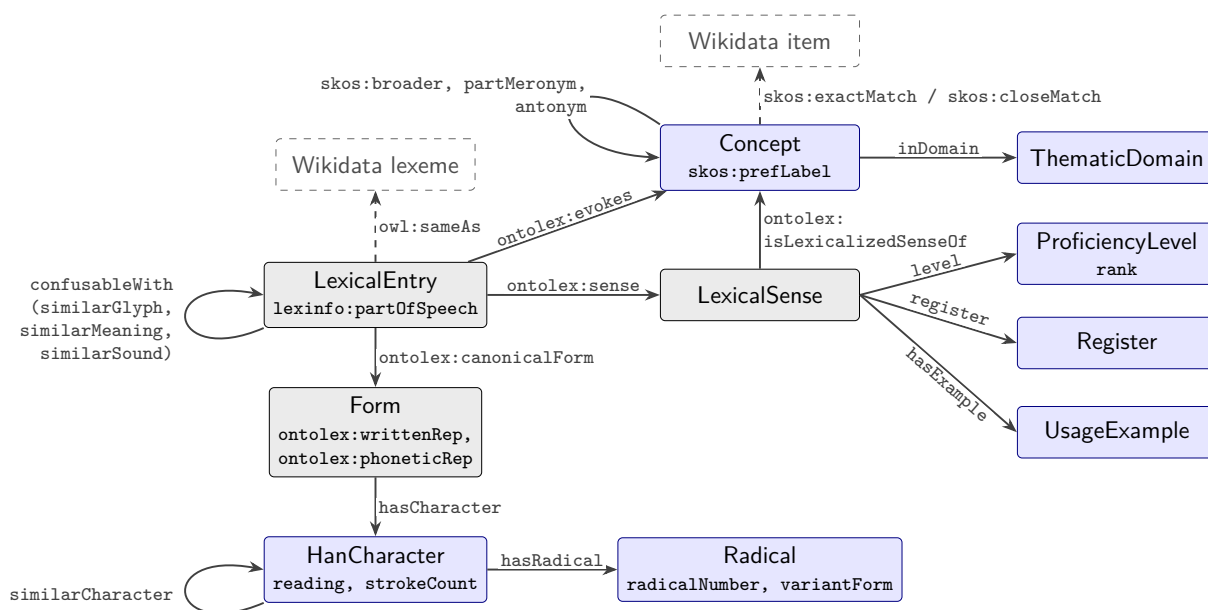


Figure 1: Overview of the LexNet ontology. OntoLex classes are grey and LexNet classes are blue. Properties without a prefix are LexNet's own.

- CQ7 Which characters make up a Chinese word, and what are the radical and the readings of each?
- CQ8 Which words contain a character with a given radical, and what do they mean?
- CQ9 Which Wikidata items and lexemes are linked to the concepts and words?

2.2 Concepts, Words and Senses

A `lexnet:Concept` is a language-independent meaning such as ‘cat’. It specialises `ontolex:LexicalConcept`, which OntoLex defines as a kind of `skos:Concept`. Each concept has one preferred label per language. A concept whose English label is ambiguous, such as either meaning of *bank*, also has an English definition (`skos:definition`). Concepts are related by `skos:broader` for hypernymy, by `lexnet:partMeronym` and its inverse `lexnet:partHolonym` for parts, and by the symmetric `lexnet:antonym`. Following Kroeger [21, §6.2.2], antonymy covers complementary pairs (‘man’ and ‘woman’), gradable antonyms (‘hot’ and ‘cold’), converse pairs (‘parent’ and ‘child’) and reverse pairs (‘come’ and ‘go’). Each concept also belongs to one topic (`lexnet:ThematicDomain`) through `lexnet:inDomain`, and the topics form a SKOS concept scheme. Lex-Info also defines antonymy and part relations, but only between senses [9]. With them, each pair would have to be stated again for every language. Since LexNet’s relations and topics are attached to concepts, each is stated once and applies to the words of all three languages.

LexNet identifies a word (`ontolex:LexicalEntry`) by its language, written form and part of speech. *Drink* as a noun and as a verb are therefore two words. OntoLex also separates entries by etymology [10]. LexNet does not, so homonyms such as *bank* ‘financial institution’ and *bank* ‘land beside a river’ form one word, as they do in the Wikidata lexeme L3354. The written form and the pronunciation belong to an `ontolex:Form`. A BCP 47 language tag [30] shows which transcription a pronunciation uses. English forms have General American IPA (“bæŋk”@en-US-fonipa) and Chinese forms have pinyin with tone marks (“māo”@zh-Latn-pinyin).

Vietnamese forms have none, as the learners already speak Vietnamese.

Each use of a word for a concept is an `ontolex:LexicalSense`. Two words are synonyms or translations when each has a sense of the same concept, so these relations are not stored as links between words (Section 2.5). Level and register are attached to the sense. A level can differ between the meanings of one word, as it does for the two senses of *bank* in Figure 2. It can also differ between words with the same meaning: 爸爸 ‘dad’ is HSK 1 and 父亲 ‘father’ is HSK 4. Register varies in the same two ways. *Cool* is informal only in the sense ‘excellent, fashionable’, and 爸爸 is informal while 父亲 is not. A level or register attached to the word would thus be wrong for some of its meanings, and one attached to the concept would be wrong for some of its words. Usage examples are attached to senses as well (Section 2.3).

Instead of a sense node, RDF-star [15] could annotate the triple that links a word to a concept. Since RDF-star is not part of OWL 2, a reasoner could not use those annotations. The sense node is standard OntoLex and keeps the data within OWL.

```
entry:en_bank_noun a ontolex:LexicalEntry ;
  lexinfo:partOfSpeech lexinfo:noun ;
  ontolex:canonicalForm form:en_bank_noun ;
  ontolex:sense sense:en_bank_noun_bank_finance , sense:en_bank_noun_bank_river ;
  ontolex:evokes concept:bank_finance , concept:bank_river ;
  owl:sameAs wd:L3354 .
sense:en_bank_noun_bank_finance a ontolex:LexicalSense ;
  ontolex:isLexicalizedSenseOf concept:bank_finance ;
  lexnet:level level:CEFR_A1 .
sense:en_bank_noun_bank_river a ontolex:LexicalSense ;
  ontolex:isLexicalizedSenseOf concept:bank_river ;
  lexnet:level level:CEFR_B1 .
form:en_bank_noun a ontolex:Form ;
  ontolex:writtenRep "bank"@en ;
  ontolex:phoneticRep "bæŋk"@en-US-fonipa .
```

Figure 2: The word *bank* in LexNet, without its usage examples. The prefixes `entry:`, `sense:`, `form:`, `concept:` and `level:` abbreviate LexNet’s hash IRIs.

2.3 Learner Layer

The learner layer covers levels, register, usage examples and confusable pairs. All four are recorded for English and Chinese, the two languages being learned. Register is recorded for Vietnamese as well. Proficiency levels are SKOS concepts in two schemes, CEFR for English and HSK 3.0 for Chinese. Each level has a numeric `lexnet:rank` that queries can compare. Every level comes from the Oxford list or the HSK syllabus, and a word missing from its list has no level. A listed level applies to all senses of a word unless the list gives them separate levels, as the Oxford list does for *bank*.

Register takes one of four values (formal, literary, informal and polite) and is recorded only for senses that differ from neutral usage. LexInfo also has a register property [9]. Its values lack informal, literary and polite. LexNet therefore defines its own property and values. For English and Chinese, the labels of the Oxford Learner’s Dictionaries [29] and CC-CEDICT [7] were consulted. The HSK Standard Course also treats 食 and 饮 as the written-language counterparts of 吃 ‘eat’ and 喝 ‘drink’ [18, p. 6], so both are recorded as literary. The values that none of the three sources covers, including all Vietnamese ones, were assigned by the team, and the reviewers checked every value (Section 3.2).

Usage examples (`lexnet:UsageExample`) are attached to senses through `lexnet:hasExample`, so each one illustrates a single meaning. They are written in the language of the word.

Confusable pairs link words rather than senses. `lexnet:confusableWith` is symmetric and has three symmetric subproperties. `lexnet:similarMeaning` links two words of one language whose meanings are close. A pair qualifies when another LexNet language colexifies the two meanings, that is, expresses both with one word [12]. It also qualifies when a learner's reference contrasts the two words. *Borrow* and *lend* meet both conditions, since Chinese uses 借 for both [7] and the Oxford Learner's Dictionaries note that they are often confused [29, entry *borrow*]. `lexnet:similarGlyph` links words that look alike, such as 八 'eight' and 人 'person'. `lexnet:similarSound` links words of one language with the same pronunciation, such as *sea* and *see* or 心 'heart' and 新 'new'. In Chinese the tone must also match. Homophone pairs are derived when the data is built, because OWL 2 DL cannot infer from two equal pronunciations that two different words sound alike.

2.4 Script Layer

A Chinese form is linked by `lexnet:hasCharacter` to `lexnet:HanCharacter` nodes. Each character has a stroke count and all of its standard readings, with the most common one marked by `lexnet:commonReading`. The reading can depend on the word: 行 is *háng* in 银行 'bank' but *xíng* in 自行车 'bicycle'. The word's pinyin therefore shows which reading applies.

Unihan [31] classifies a character under a radical, usually one of its components. 妈 'mother', for example, contains 女 'woman' and 马 'horse' and is classified under 女. LexNet records only this radical (`lexnet:Radical`) and does not link 妈 to 马. A few characters, such as 书 'book', have two classifications in Unihan. LexNet takes the first, so every character has exactly one radical. A radical has a Kangxi number and, where one exists, a common variant form (𠂇 for radical 94). A simplified form, such as 𠂇 for radical 120 in 红 'red', is a separate radical, as in Unihan. Because every character of a word is linked, a compound can be found through the radical of any of its characters. 狮子 'lion' and 猴子 'monkey' thus share radical 94 with 猫 'cat' through their first characters.

Look-alike characters are linked directly by the symmetric `lexnet:similarCharacter` rather than through words. Many characters in LexNet occur only inside compounds, such as 午 'noon' in 上午 'morning' and 午饭 'lunch'. One statement that 午 and 牛 'cow' look alike therefore covers every word that contains either character. 买 'buy' and 卖 'sell' are both characters and words, so they are linked by `lexnet:similarCharacter` and by `lexnet:similarGlyph`. `lexnet:similarCharacter` is not a subproperty of `lexnet:confusableWith`, whose domain is words. As a subproperty, it would make the reasoner infer that a character is a word, which the disjointness axioms (Section 2.5) forbid.

2.5 Axioms and Reasoning

Table 1 summarises the main axioms of the ontology. The first group lets a reasoner detect errors in the data (Section 6). The second lets it derive facts that the data need not state.

OWL makes no unique-name assumption, so the functional properties alone would not catch a sense with two registers. The reasoner would only conclude that the two registers are the same individual. The all-different axioms turn such a case into a contradiction. Omissions, in contrast, are not detected. Under the open-world assumption, a word without a pronunciation is not an error, and an axiom that required one would only make the reasoner assume an unknown

Table 1: Main axioms of the LexNet ontology. Outside the formulas, properties without a prefix are LexNet’s own.

Axiom	Terms or formula	Effect
<i>Detecting errors</i>		
Disjoint classes	the ten classes of Figure 1	a node in two of them is a contradiction
Domain and range	most LexNet properties, e.g. register from senses to registers	with disjointness, a link to the wrong type of node is a contradiction
Functional	level , register , hasRadical	at most one value per node
Cardinality 1	ontolex:isLexicalizedSenseOf	exactly one concept per sense, stricter than OntoLex
All different	concepts, levels, registers, radicals	a second concept, level, register or radical is a contradiction
<i>Deriving facts</i>		
Defined class	$PolysemousEntry \equiv LexicalEntry \sqcap \geq 2 \text{ sense.LexicalSense}$	polysemous words, homonyms included
Property chain	$sense \circ isLexicalizedSenseOf \sqsubseteq evokes$ $evokes \circ evokes^- \sqsubseteq sharesConceptWith$	the concepts a word evokes synonyms, translations and the word itself
Symmetric	antonym , similarCharacter , confusableWith and its subproperties	the reverse of each pair
Subproperty	similarGlyph , similarMeaning and similarSound	the parent relation confusableWith
Inverse	partMeronym and partHolonym	the inverse of each part relation

value. The build script checks completeness instead (Section 3.2), as SHACL shapes [20] could also do.

`lexnet:PolysemousEntry` removes the need to tag polysemous words by hand. The reasoner can count two senses because each one lexicalises exactly one concept and the concepts are all different. The class counts senses rather than evoked concepts, because OntoLex defines `ontolex:evokes` by a property chain and OWL 2 DL allows no cardinality restriction on such a property. The SPARQL endpoint does not reason, so `ontolex:evokes` is also stated in the data, and the queries compute the other derived facts they need (Section 5.1).

The reused terms are declared with the same entity types as in the official OntoLex, SKOS and LexInfo files rather than imported, which keeps the ontology small. The ontology file, which also contains the levels, registers, parts of speech and concept schemes, has 139 logical axioms, 15 classes, 25 object properties and 9 datatype properties. Its expressivity is $\mathcal{ALCRQIQ}(\mathcal{D})$. Every class and every object and datatype property has an English and a Vietnamese label, so an interface can switch language using the data alone.

3. Data Collection and RDF Transformation

The vocabulary of LexNet was entered and reviewed in a shared spreadsheet. Two scripts turn it into RDF: one exports the spreadsheet to CSV files, and the build script converts these files, the ontology and the character data into a single RDF file in Turtle syntax (Figure 3).

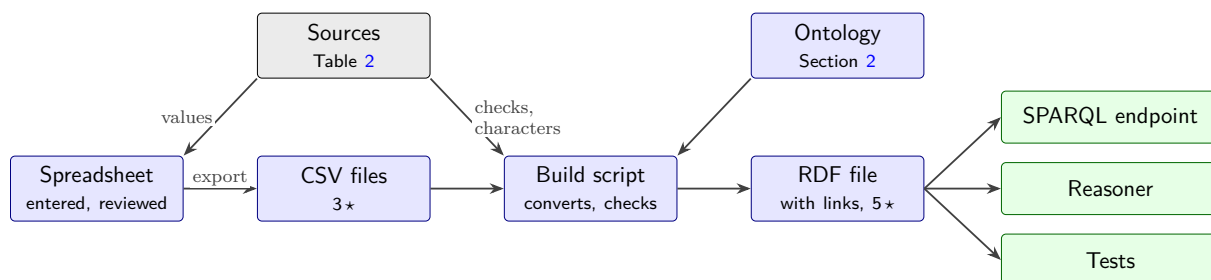


Figure 3: From the spreadsheet to the RDF file. Blue boxes are LexNet’s data, ontology and build script, the grey box stands for the sources, and green boxes show how the RDF file is used.

3.1 Scope and Sources

LexNet contains 236 concepts, expressed by 759 words with 774 senses in the three languages. The concepts belong to 11 topics: animals, the body, family and people, food and drink, colours, nature and weather, time, numbers, places and things, actions, and qualities. Candidate concepts were taken from the words of the Oxford list and the HSK syllabus. A candidate was kept if it had a word in each of the three languages. At least one of these words also had to be at CEFR B2 or below in English or at HSK 6 or below in Chinese. Two concepts miss this condition but are kept for their relations: *mammal* links *animal* to kinds such as *cat*, and *claw* is a part of *cat*. Only nouns, verbs, adjectives and numerals are included. Function words are left out, as they rarely have a counterpart in each of the three languages. English articles, for instance, have none in Chinese.

The three languages do not always divide meanings in the same way. Vietnamese uses *trẻ em* for a young person and *con* for a son or daughter, whereas English *child* [29, entry *child*] and Chinese 孩子 cover both meanings. With a single concept, *trẻ em* and *con* would become synonyms (Section 2.2). LexNet therefore has two concepts. *Child* and 孩子 have a sense for each, so the reasoner classifies them as polysemous (Section 2.5).

The selection also deliberately includes examples of each feature of Section 2, such as polysemous words, synonyms that differ in register, homophones, look-alike characters and characters with several readings. Their number in LexNet does not reflect how common they are in the three languages.

Table 2 lists the sources of the data. The values taken from the two word lists are part of LexNet, but the lists themselves are not redistributed.

3.2 Collection and Review

The spreadsheet has a tab for the topics, one for the concepts and their relations, one per language with a row for each sense, and one for the confusable pairs. One team member created the rows from the selected candidates and added the Vietnamese words. Values that the sources provide were pre-filled: levels from the two word lists, pinyin from the HSK syllabus, IPA from the CMU dictionary, and Wikidata items and lexemes suggested by a script (Section 4).

Two other team members then reviewed every concept, sense and confusable pair and marked each as checked. Where needed, they corrected or added words, relations, levels, registers and Wikidata links. They also completed the usage examples, one for each English and Chinese sense. The examples were written rather than copied from dictionaries, so that they can be

Table 2: Sources of the data.

Source	Used for	Licence or use
Oxford 3000 and 5000 [28]	candidates; CEFR levels, per sense where the list separates senses	not redistributed
HSK 3.0 syllabus [8]	candidates; HSK levels and pinyin	not redistributed
Oxford Learner’s Dictionaries [29], CC-CEDICT [7], HSK Standard Course [17]	meanings, register labels, confusable pairs	consulted only
Princeton WordNet [25]	hypernymy and part relations	consulted only
CMU Pronouncing Dictionary [6]	English IPA	BSD 2-Clause
UniHan [31]	radicals, readings and definitions	Unicode License v3
Make Me a Hanzi [19]	stroke counts	Arphic Public License
Wikidata [32]	links to items and lexemes (Section 4)	CC0

released under an open licence.

Recurring cases were decided once and recorded as rules. One of these rules concerns synonyms, which must be interchangeable in the meaning of their concept. Vietnamese and Chinese usually distinguish an older from a younger brother (*anh trai* and *em trai*, 哥哥 and 弟弟). Unlike the two meanings of *child* (Section 3.1), these are kinds of brother. Listing them as words of the concept *brother* would make them synonyms, so the concept has the general words *anh em trai* and 兄弟 instead. Regional variants, in contrast, are interchangeable, so Vietnamese *lợn* and *heo* ‘pig’ are synonyms. They have no register, since LexNet’s registers describe formality and style, not region.

The build script adds automatic checks to the review. Whenever it runs, it reports values that differ from the word lists or the CMU dictionary, pinyin that matches no reading of its characters, and missing words, levels or pronunciations (Section 2.5). All warnings were resolved in the spreadsheet, and the final data produces none.

3.3 RDF Transformation

The CSV files are published in the project’s repository under the CC BY 4.0 licence. As openly licensed, structured data in a non-proprietary format, they reach three stars [1], as marked in Figure 3. The RDF file built from them contains the ontology followed by the data, so the SPARQL endpoint and the reasoner need to load only one file. The ontology header also declares the licence with `dct:license`.

Some data in the RDF file does not come from the CSV files. Each Chinese word is linked to its characters, whose radicals, readings and stroke counts come from the character data (Table 2). The build script also derives the homophone pairs (Section 2.3), states `ontolex:evokes` for the SPARQL endpoint (Section 2.5) and adds the all-different axioms for concepts and radicals. These axioms depend on the data and are thus not in the ontology file. The whole file has 15,646 triples, with 286 characters under 116 radicals.

The fourth star requires IRIs that identify things [1]. LexNet uses hash IRIs. A data node is named by the address of the RDF file followed by # and a local name, and LexNet’s own terms, such as its classes and levels, by the address of the ontology file in the same way. A concept is named by an English word, with a keyword where the word is ambiguous (`concept:bank_river`). A word is named by its language, written form and part of speech (`entry:en_bank_noun`), and a

sense adds the concept to the name of its word (Figure 2). Characters and radicals are named by the character itself and by the radical number (`char:猫`, `radical:94`). Since the part after `#` is not sent to the server, looking up any data IRI returns the whole RDF file. A static web host can therefore make every LexNet IRI dereferenceable (Section 5). Links from LexNet’s concepts and words to Wikidata add the fifth star (Section 4).

4. Linking to Wikidata

LexNet links its concepts to Wikidata items and its words to Wikidata lexemes (Figure 1). Because a concept is shared by the three languages, a link on the concept connects the words of all three to an item. The links give access to what Wikidata records about each meaning and word, such as labels in further languages and recorded pronunciations (Section 5).

4.1 Link Properties

Concepts are linked with `skos:exactMatch`, or with `skos:closeMatch` when the item is only close in meaning [24]. They are not linked with `owl:sameAs`, which would make a concept and an item one individual with the same statements. A LexNet concept is the meaning of words, whereas an item describes the thing itself. Item Q146 ‘cat’, for example, is a subclass of *Felidae*, which the meaning ‘cat’ is not. Identity would also merge concepts linked to the same item. Both meanings of *old* are linked to item Q12943371, ‘not young’ by `skos:exactMatch` and ‘not new’ by `skos:closeMatch`. With `owl:sameAs`, the reasoner would conclude that the two meanings are one concept, which contradicts the all-different axiom (Section 2.5).

Words are linked with `owl:sameAs`, as *bank* is to L3354 in Figure 2. Unlike an item, a lexeme is a word, which Wikidata describes in RDF as an `ontolex:LexicalEntry` with a lemma, a language and a lexical category [34]. LexNet identifies a word by the same three properties (Section 2.2), so a word and the matching lexeme are the same word.

The three properties do not always lead to exactly one lexeme. Wikidata sometimes separates homonyms by origin, as *OntoLex* does [10]. It has four English noun lexemes *plane*, and LexNet’s *plane*, which has only the meaning ‘aircraft’, is linked to the one with this meaning, L46157. Other words have a lexeme with the same lemma and language but another category. Wikidata classifies some English cardinal numbers, such as *two*, as determiners, while the Oxford list has them as numerals. Similarly, 今天 ‘today’ and 明天 ‘tomorrow’ are adverbs in Wikidata but nouns in the HSK syllabus. These words are still linked, because a sense of each lexeme denotes the item of the word’s concept (property P5137). A word and its lexeme are thus one word that the two resources classify differently, not two words with different meanings like the noun and the verb *drink*.

4.2 Link Discovery

Several Wikidata items share the English label *cat*, so a search by label returns unrelated candidates. A script instead starts from Wikidata’s lexical data. For each concept, it takes the English lexemes with the concept’s English label and part of speech and suggests the items that their senses denote (P5137), together with the glosses of these senses. For each word, it suggests the lexemes that share its three properties (Section 4.1), accepting determiners for numerals. The script fills in a link only when there is a single suggestion and, for a concept, the English word is not polysemous in LexNet.

The other links were chosen by meaning, from the suggestions or through a search on Wikidata. ‘Potato’, for example, belongs to food and is linked to the staple food (Q16587531) rather than to the plant species (Q10998). For *sit* and *stand*, the closest items are the postures ‘sitting’ and ‘standing’, which are linked as close matches. The reviewers then checked every link manually (Section 3.2).

4.3 Coverage

Of the 236 concepts, 235 are linked to an item, with 230 exact and 6 close matches. *Person* has one of each, and no matching item was found for *cool* in the sense ‘pleasantly cold’. Words are covered unevenly (Table 3). Every English word is linked to a lexeme, but fewer than half of the Chinese words and only 6 Vietnamese words are. For the other words, Wikidata has no lexeme with the same three properties, so the gaps reflect its coverage, not the matching. The context that the links add is uneven in the same way, as recordings and descriptions are rarer in Chinese and Vietnamese than in English. LexNet therefore keeps its own pronunciations and uses Wikidata only for additional context.

Table 3: Coverage of the links to Wikidata by language. Recordings and descriptions were counted in October 2026.

	English	Vietnamese	Chinese
Words	237	256	266
linked to a lexeme	237	6	118
whose lexeme has a recording	183	1	7
Distinct linked items with a description (of 234)	234	129	163

5. Data Access

The project’s repository (<https://github.com/Nov17th/lexnet>) contains the ontology, the CSV files, the build script, the queries and the web application. GitHub Pages serves the RDF file¹ and the ontology file² at the addresses on which their IRIs are based (Section 3.3), with the media type `text/turtle`, so every LexNet IRI can be looked up as Linked Data [3]. The data can also be queried and browsed locally, at a SPARQL endpoint and in a web application.

5.1 SPARQL Endpoint and Queries

The SPARQL endpoint is Apache Jena Fuseki [2], which loads the RDF file and accepts only read queries. The repository contains SPARQL 1.1 queries [14] for all competency questions of Section 2.1, and Table 4 shows how they answer each one.

Several of these features stand in for reasoning, which the endpoint does not perform (Section 2.5). Instead of `lexnet:sharesConceptWith`, which a reasoner would derive from a property chain, the query in Figure 4 follows the same chain as the path `ontolex:evokes/~ontolex:evokes` and finds *cat* and *mèo* from 猫. The query for polysemous words counts senses instead of using the class `lexnet:PolysemousEntry`. Antonyms are read in both directions, because the data states some pairs in one direction only. Confusable words are found by listing the three subproperties of `lexnet:confusableWith` with VALUES, since the data states only the subproperties.

¹<https://nov17th.github.io/lexnet/build/lexnet-full.ttl>

²<https://nov17th.github.io/lexnet/ontology/lexnet-ontology.ttl>

Table 4: How the queries in the repository answer the competency questions. Property prefixes are omitted.

Question	Main SPARQL features
CQ1	join from the concept through the senses to the words and levels, property path evokes/ˆevokes
CQ2	two words of one concept, FILTER on the same language, OPTIONAL register
CQ3	GROUP BY with HAVING (COUNT(?x) >= 2) , OPTIONAL level, register and example
CQ4	broader+ for kinds, partMeronym for parts, antonym ˆantonym for opposites
CQ5	FILTER on the numeric rank of a level within a topic
CQ6	VALUES over the subproperties of confusableWith , GROUP BY pronunciation, similarCharacter ˆsimilarCharacter
CQ7	hasCharacter , GROUP_CONCAT of readings, character order with STRBEFORE
CQ8	VALUES over radicals, EXISTS with broader*
CQ9	SERVICE to Wikidata through exactMatch and sameAs

```
PREFIX ontollex: <http://www.w3.org/ns/lemon/ontollex#>
SELECT DISTINCT ?word WHERE {
  ?e1 ontollex:canonicalForm/ontollex:writtenRep "猫"@zh .
  ?e1 ontollex:evokes/ˆontollex:evokes ?e2 .
  FILTER(?e1 != ?e2)
  ?e2 ontollex:canonicalForm/ontollex:writtenRep ?word .
}
```

Figure 4: A query that returns the words that share a concept with 猫.

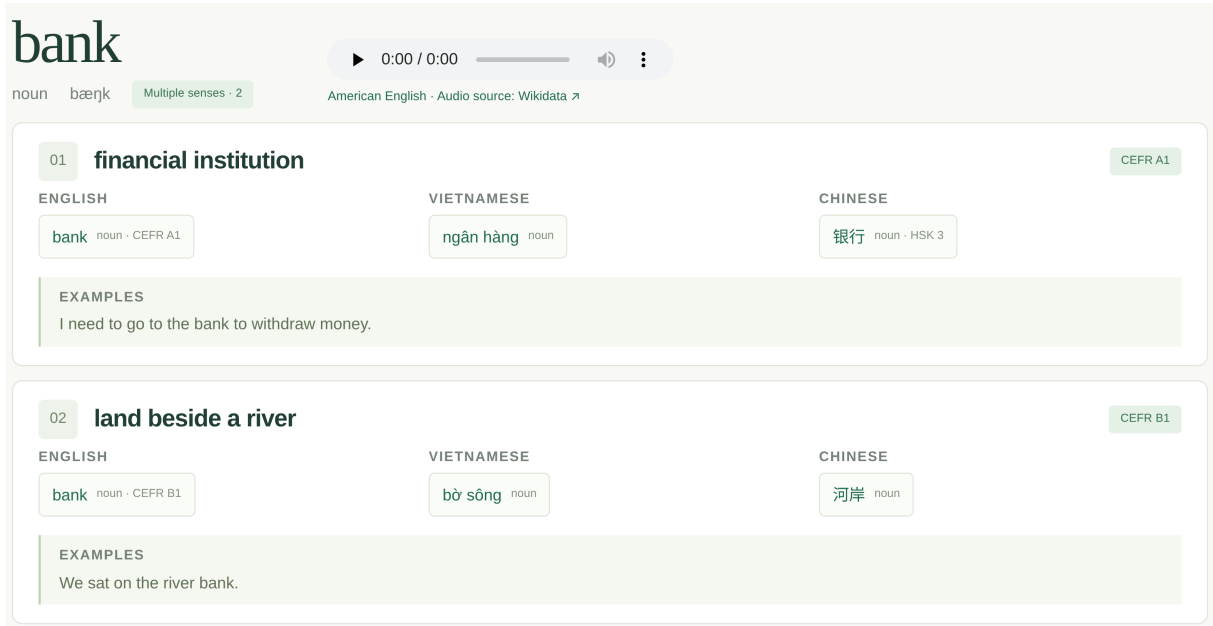
The three queries for CQ9 add data from the Wikidata endpoint (<https://query.wikidata.org/sparql>) through **SERVICE**. One follows the **skos:exactMatch** link of ‘cat’ and returns its Wikidata labels in seven languages, four more than LexNet has. Another follows **owl:sameAs** from English words to their lexemes and shows the IPA and recordings of Wikidata next to LexNet’s IPA. The third compares LexNet’s pinyin with that of the linked Chinese lexemes.

5.2 Web Application

The web application presents the data to learners and developers. Its Next.js server sends SPARQL queries to Fuseki and passes the results to the browser, so all data that the application shows comes from the RDF file, without a database of its own. The recordings are the exception: the server fetches them from the Wikidata lexemes linked to the words.

Learners can search for words in the three languages. The page of a word shows its part of speech, pronunciation and senses (Figure 5). Each sense shows the level, register and usage example recorded for it, and the words that share its concept serve as translations and synonyms. Links lead to related concepts, such as kinds, parts and opposites, and from Chinese words to their characters, with radicals, readings and look-alike characters. Confusable words are listed by the type of confusion, and a topic page lists the words of a topic up to a chosen level.

For developers, a SPARQL console runs the repository’s queries and any other read query, and a graph view shows the resources around a word as nodes and edges. Both expose the RDF behind the learner pages.



bank 0:00 / 0:00 Multiple senses - 2 American English - Audio source: Wikidata

01 financial institution CEFR A1

ENGLISH	VIETNAMESE	CHINESE
bank noun - CEFR A1	ngân hàng noun	银行 noun - HSK 3

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I need to go to the bank to withdraw money.

02 land beside a river CEFR B1

ENGLISH	VIETNAMESE	CHINESE
bank noun - CEFR B1	bờ sông noun	河岸 noun

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We sat on the river bank.

Figure 5: Excerpt of the page for *bank* in the web application. Each sense has its own level, translations and example, and the recording comes from the linked Wikidata lexeme.

6. Evaluation

The profile checker of the OWL API [16] finds both the ontology file and the RDF file within OWL 2 DL. Neither fits any of the profiles EL, QL and RL [33], which allow simpler reasoning. RL fails only on the class `lexnet:PolysemousEntry` and two restrictions to exactly one value, but these include the definition of polysemous words and the restriction of a sense to one concept. LexNet therefore needs a reasoner for full OWL 2 DL, such as HermiT [13].

6.1 Reasoning

HermiT finds the data consistent and infers what the axioms are meant to derive. It classifies the 15 words with several senses as polysemous, relates the 20 confusable word pairs by `lexnet:confusableWith` in both directions and infers the reverse of the 16 pairs of look-alike characters. Spot checks find no unwanted inference: a character is not classified as a word, and a word with one sense is not classified as polysemous.

To check that the axioms also detect errors in the data (Section 2.5), the test script adds one wrong triple at a time, such as a second register for a sense, a level that points to a concept or antonymy between two words, and asks HermiT whether the data is still consistent. The triples are built from nodes that queries select, so the test keeps working when the data changes. HermiT detects all 12 faults through the axioms of the first group in Table 1, and the test script can repeat this check and those above whenever the RDF file is rebuilt (Figure 3).

6.2 Radicals and Meaning

Li et al. [22] note that the radical 犳 refers to mammals (Section 1). The query for CQ8 tests this on LexNet’s vocabulary. All five words with a character under 犳 name mammals: 狗 ‘dog’, 狮子 ‘lion’, 猪 ‘pig’, 猫 ‘cat’ and 猴子 ‘monkey’. Six of the nine words with a character

under 水 ‘water’, written 氵 in most of them, concern water or another liquid, such as 河 ‘river’ and 果汁 ‘juice’. 水果 ‘fruit’, 汽车 ‘car’ and 温度 ‘temperature’ do not. A radical is thus a useful hint to the meaning of a word, but not a rule.

7. Conclusion

LexNet organises 759 English, Vietnamese and Chinese words around 236 shared concepts. Translations and synonyms follow from the concepts that the senses point to. English *sugar* thus reaches Chinese 糖 through their shared concept, not through Vietnamese *đường*, which would also lead to 路 ‘road’ (Section 1). Levels, registers and usage examples are attached to senses, and Chinese words are broken down into characters and radicals. The data is published as one RDF file under an open licence, with dereferenceable IRIs and links to Wikidata items and lexemes, so it is five-star Linked Open Data. A SPARQL endpoint answers the competency questions, a web application presents the data to learners, and a test script checks the data against the ontology with HermiT.

The vocabulary, however, covers everyday topics only, and the numbers of polysemous words, homophones and similar cases reflect the selection rather than the three languages (Section 3.1). Vietnamese and Chinese words gain less from the links than English ones, because Wikidata has fewer lexemes and recordings for them and fewer descriptions in their languages (Section 4.3). The SPARQL endpoint and the web application run locally rather than as public services.

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